Requirements of People with Disabilities and Caregivers for Robotics: A Case Study

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Abstract. Robotics offers new solutions for digital customer interaction. Social robots can be used in applications such as customer support, guiding people to a location on company premises, or entertainment and education. An emerging area of research is the application in community facilities for people with disabilities. Such facilities face a shortage of skilled workers that could be addressed by robotics. In this work, the application of social and collaborative robots in care facilities and workshops for the disabled is presented by providing a requirements analysis. The use of the humanoid robot Pepper in assisted living was tested and subsequently evaluated in interviews with caregivers who initiated and observed the interaction between the group and the robot. Additionally, robotic applications in assisted work were assessed, resulting in a divergence from the industrial use of robots. A comparative overview with recent literature is presented. The connection between the community home and the workshop raised the question of whether the use of different robots in both places could lead to conflicts.

Keywords: Assistive Robotics · Assisted Living · User Experience · Social Robots

1 Introduction

An ongoing trend is the use of robotics in social and medical environments. Early versions of social robots are represented in the early 2000s with toys such as Furby (Tiger Electronics) or Aibo (Sony) that change their behavior depending on the user's behavior [2]. Today, social robots are motivated by new challenges such as the prevention of loneliness and social isolation as seen in the Covid-19

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pandemic [8,13], assistance in activities of daily living (ADL) through cognitive and physical stimulation [13], and human-friendly navigation to avoid collisions in logistic tasks, but also in facility tours to present the location to customers [18]. In healthcare, they are used to assist healthcare workers by reducing time-consuming tasks such as picking up and delivering bed sheets and other supplies, allowing caregivers to focus on patient needs and more critical tasks [10]. As a result, the market and research interest in social robots is growing [1,15].

The application of social robots varies greatly depending on the audience and purpose. According to the IFR (International Federation of Robotics), they are categorized as a subgroup of service robotics, as opposed to industrial robotics [5]. Service robots are designed to perform human-centered tasks outside of industrial automation applications [6]. A division into social and non-social robots and domestic or professional use further defines the application of the robot [12]. For this reason, design focuses not only on functionality, but also on form and material to increase acceptance [13]. This can be seen in the diverse field of social robotics, from humanoid robots such as Pepper and NAO (Aldebaran United Robotics Groups) [10], to pet-like companion robots like Spot (Boston Dynamics) and PARO (PARORobots USA), to service robots used in health and social care facilities to optimize logistics and reduce time pressure on caregivers such as Moxi Robot ([10]) and Relay (Swisslog Healthcare).

In contrast, assistive robots, as they are colloquially known, are used for rehabilitation or assistance in everyday life. They do not have the ability to speak and are mostly based on robotic arms, but still need adaptive capabilities. Rehabilitation robots have the ability to guide the user's extremities to assist in the rehabilitation process [10]. The robots' force feedback allows progress to be tracked and adapted to each individual patient. The gamification of such processes is being explored through the implementation of computer games or virtual reality [10]. Assistive Robotic Arms (ARA) can be attached to wheelchairs to assist in performing daily tasks through a joystick, head tracking, eye tracking, or brain computer interface (BCI). The control of such systems can be facilitated by shared control that automates task execution for example by using intention recognition [22].

With the ability to recognize emotions and personality and store memories of previous interactions, such robotic systems can benefit care facilities and hospitals [1]. Adaptive behavioral models, cognitive architectures design, and the ability for perceived empathy in human machine interaction are recent challenges. The application of artificial intelligence accelerates the solution finding [15].

These challenges and requirements must be evaluated separately for each individual robot and target group (e.g., customers, elderly people or people with disabilities). In this work, the user requirements have been identified in cooperation with two related organizations. A case study is presented in which requirements and solutions for assisted work were identified and the robot Pepper was evaluated in an assisted living facility. A comparative overview with existing literature was performed to validate the results.

2 Methods

This work is based on a cooperation with Lebenshilfe im Kinzig- und Elztal e.V. and the company WfB Haslach gGmbH. Lebenshilfe im Kinzig- und Elztal e.V. is an organization that provides services for people with disabilities to ensure participation in all areas of life and to maintain a good quality of life. Services include counseling for family members, assistance in finding housing, suitable employment, and leisure activities [11]. Lebenshilfe im Kinzig- und Elztal e.V. is the shareholder of WfB Haslach gGmbH. This company offers people with disabilities a wide range of work and employment opportunities that are adapted to each individual's abilities [21].

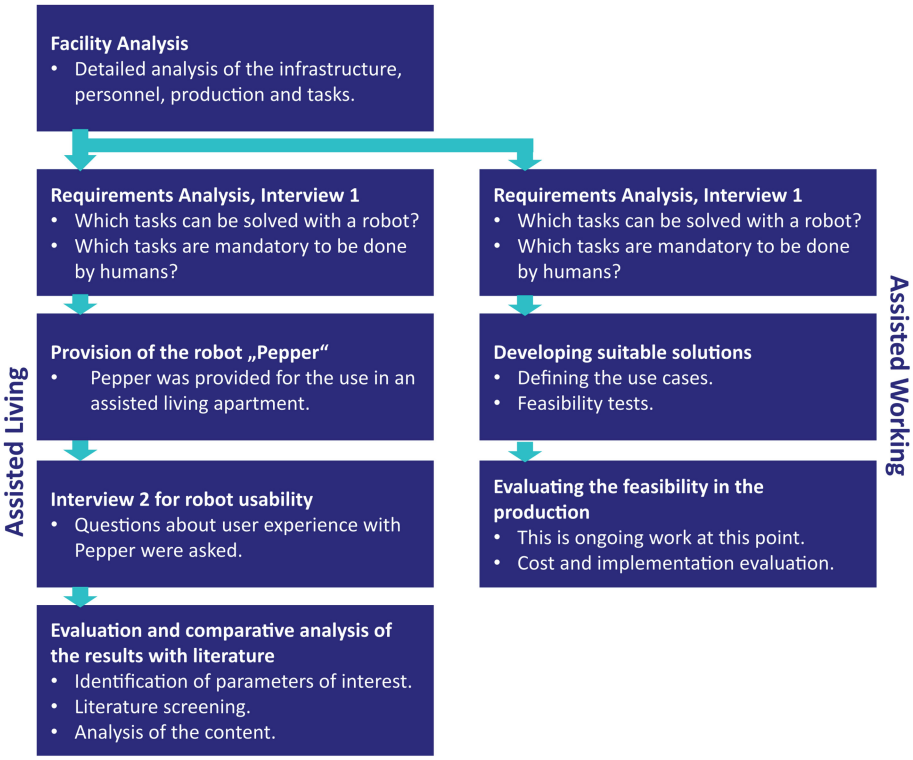

Fig. 1. Application and requirement analysis process.

In a multi-step process, as shown in Fig. 1, the authors identified the needs of both institutions with respect to robotic applications. This work took place between March and October 2023. After an on-site analysis, the evaluation was split into two requirements analyses due to the different needs and work areas. While in the assisted work a demand was found to relieve the caregivers by reducing quality control in production, in assisted living entertainment and assistance

in cleaning and logistics were identified, which will be presented in detail in the following sections.

The robot Pepper was used to evaluate the user experience in assisted living (Fig. 3). The robot is 1.20 m tall and weighs 28 kg. The battery life is 12 h according to the manufacturer. Pepper was equipped with applications such as games (memory, guessing, and animal imitation) and interaction options. The interactions were limited to hugging people, speaking fixed scripts, posing for selfies, and recording greetings to relatives or friends. The robot is able to locate itself and move to predetermined locations. A speech recognition tool provides the robot with an additional input modality to the touch screen.

3 Results

The analysis began with a visit to both facilities. An on-site evaluation of both facilities was conducted. The authors spoke with management and staff. A standardized questionnaire was not used. The requirements of each facility are presented below.

3.1 Assisted Living

The community housing complex visited is divided into three communal apartments, each with space for 10 residents (Fig. 2). The residents are adults and pensioners. Most of the residents work at WfB Haslach gGmbH. People with different disabilities live together. They are cared for around the clock by experienced caregivers, but they do most of the daily tasks themselves, such as cooking together and keeping their rooms clean. To ensure equality for people with disabilities the German government supports housings like this. One goal is to allow people with disabilities to choose the type of housing that best suits their abilities and preferences. Another goal is to decentralize the complexes, which will lead to less segregated and closed living environments [14]. The modern complex represents this agenda. Since every part of the house is accessible, it provides perfect conditions for social, mobile robots such as the humanoid robot Pepper.

The requirement analysis was conducted in the steps mentioned in Fig. 1. First, caregivers and managers were asked, what features they thought the optimal robot should have and what tasks the robot would be most helpful with. The results showed that the robot would be most helpful if it could guide people to their rooms, retrieve objects from other rooms, such as a glass of water from the kitchen, assist with eating, or provide general entertainment functions such as playing a game. Due to the Covid-19 pandemic, a feature that regularly cleans surfaces and door handles according to the assisted living standards would be helpful. When walking, some residents may need assistance due to the risk of falling, or they may walk slower than the rest of the group, in which the robot could overtake the role of an additional caregiver to walk with them at their own pace. On the other hand, the robot should not access areas such as

the bathroom due to privacy restrictions, nor should it distribute medication. Although the robot could help by monitoring and assisting in cleaning each resident's room, the management pointed out that this would raise further ethical questions about privacy regulations. Due to the infrastructure of the building, it was also requested that the robot could be able to ride the elevator or go outside the building to assist with walking activities.

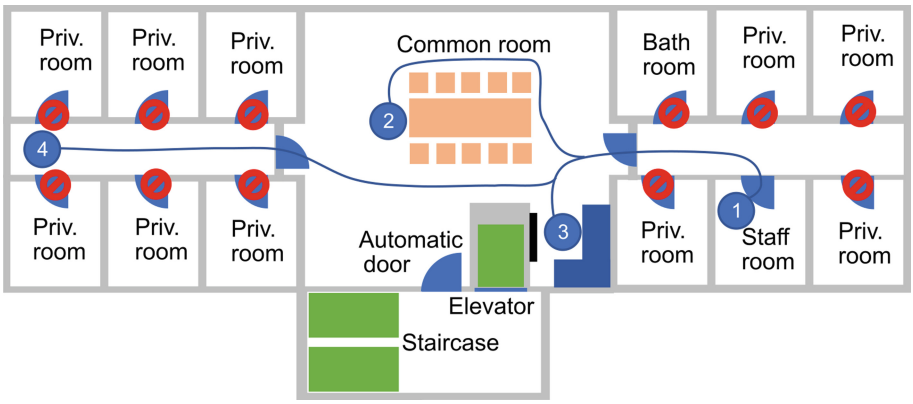

Fig. 2. Example of the shared apartment. It is accessible through an automatic door that can be opened by pressing a button. 1. Robot is loaded in this place overnight. 2. and 3. Robot location for entertainment interaction. 4. Trajectory to guide residents back to their rooms. Private rooms and the bathroom should not be accessed by the robot.

In a second step, the organization was interested in testing such a robot in the shared apartment. The Pepper robot was assigned to one floor of the building, but the residents could visit each other as the rooms are connected to a common room on each floor. Over a period of eleven days, the robot was active for at least one hour every day except for two days. Caregivers monitored the interactions and ensured a safe and privacy-compliant environment for the residents. Therefore, the robot was only used in the common areas.

In the third step, the caregivers collected the residents' experiences with Pepper and participated in the second interview. Table 1 lists the resulting feedback key points from residents and caregivers.

The setup took approximately two hours at the facility. During this time, the robot's functions were explained and a path was taught from the loading cable in the nurses' staff room to the common room, as shown in Fig. 2. Although all doors are accessible and at least 1 m wide, Pepper had difficulty maneuvering between rooms due to inaccurate positioning. While teaching the robot, instructions were displayed on the touchscreen. Especially in path-teaching mode, the robot had no artificial feedback, such as a sound or flashing LEDs, to signal that it had completed its task. Since the robot was moving away from the user

Table 1. Feedback key points regarding the usage of Pepper.

No.	Key point	Perception
1	Mood-lifting effect	positive
2	Speech abilities: discreet with personal information	positive
3	Setup time of about two hours	positive
4	Limited localization accuracy	negative
5	No artificial feedback to signal the end of a task	negative
6	Not usable in a loud group	negative
7	Switched languages unintentionally	negative
8	No pressure sensors to detect contact	negative
9	Permanent upper body movements	negative
10	Maximum volume setting was too low	negative
11	Proposal for a more stable standing position	–
12	Proposal for smart home device control	–
13	Speech model and hardware was perceived as outdated	negative
14	Proposal that robot should be able to play music	–

most of the time, this was inconvenient. Therefore, the staff suggested the possibility of a split-screen application in future systems, showing the instructions on a handheld device. Automatic movement to a loading station and additional automatic loading was requested.

In general, the residents liked Pepper. It had an uplifting effect on the residents. The reason for this, besides the entertainment functions, was that the residents could tell the robot everything without being laughed at or gossiped about with others. In the beginning, one person was against the system, but as the days went by, the acceptance improved. One reason for this was the observation of the system over several days, to see that the robot stops in front of other people when it moves and does not hurt anyone.

The usability has been criticized in some points. The robot is not usable in large groups where multiple users want to interact with the system, as it is designed to interact with one person at a time. The test environment was naturally noisy and busy which is why the robot was often unable to recognize commands. This issue is signaled by Pepper when it turns its head from one person to the next. At that moment, the attention switches to the person whose face the recognition algorithm recognizes best. In some applications, the robot unintentionally switched to English because no German version of the function was provided. Most of the residents do not speak English. Many wanted to try the handshake function. The robot has touch sensors on the back of its hands, but they were not used in the handshake function. After an internal countdown, it began shaking hands, which caused irritation. When using the Memory app, Pepper kept moving the upper body where the touchscreen is

mounted. This made it difficult to select cards. Furthermore, residents wished Pepper had more conversational capabilities. With the ongoing development in smart devices technology, residents and caregivers are accustomed to newer technology. In particular, the robot's responsiveness was outdated due to their daily use of voice assistants such as Alexa (Amazon), Siri (Apple), or Google Assistant (Google). Finally, residents mentioned that Pepper is naked. They wanted to dress him up for fun. The customization of parts could increase the acceptance of the humanoid robots, and is provided by additional manufacturers for Pepper [16].

In terms of accessibility, hearing-impaired people were unable to interact with the robot because it was too quiet. In addition, the robot needs to be build more stable because it could be kicked unintentionally since some disabilities, such as spasticity, do not allow the constant control of body movement. Another issue is coverage by private and government health care systems. The cost of a robot that exceeds a certain price cannot be covered by the institutions. This is also true if the system is to be used by individuals living on their own.

Finally, feedback was sought on specific tasks that the robot should help with after trying the robot for eleven days. Household tasks should not be done by the robot, since in this type of all-day care, the residents are responsible for themselves. Instead, the robot should perform pick and place tasks. Implementing Bluetooth and smartphone applications could help, as one of the most common requests was to play music. This would also allow bedridden residents to call the robot. One suggestion to improve the robot was that it should be able to push a wheelchair, making it easier for wheelchair dependent residents to get around.

3.2 Assisted Work

The workshops of WfB Haslach gGmbH are located in Elzach, Hausach and Steinach in Baden-Württemberg, Germany. These facilities produce various products such as printed pens and shirts, wallets, key chains, sewing kits and parts for automotive and other industries. The work can be described as predominantly assembly work. Three main differences from general production lines have been identified.

- **Plant utilization:** Utilization capacities fluctuate due to varying order situations caused by delivery bottlenecks and peak production times specified by the customer.
- **Daily quantities:** Due to the disabilities of the workers, the quantity may vary from day to day. Unlike most industries and due to the low level of automation, absent workers cannot be replaced by other workers. Most workers have a limited range of varied work in which they specialize.
- **Various production lines:** The aim of this company is to promote the employment of disabled people. Therefore, most of the work is done manually. Many machines were presented to assist the workers in production, such as sewing machines, printing machines (screen and pad printing), and devices for aligning parts and assisting with assembly tasks.

The role of the workers varies. Most of manual work is done by people with disabilities. Trained supervisors monitor the work process, assist workers in the process, but also in activities such as assisting with toileting, and perform tasks such as quality control. Due to the lack of digitalization and the limited use of high-tech control and measuring equipment, quality control is also done manually. As a result, this task has to be interrupted when another person needs assistance, resulting in a restricted workflow. This is one of the company's biggest challenges. Based on these statements, the requirements for the robotic system were identified and presented in Table 2.

The second step was to develop and evaluate solutions. Using a state-of-the-art camera, solutions were provided for quality control of various products. For example, a computer vision system was set up to verify that a print on a pen was applied correctly by color and position. The algorithms used for measurement were successful. Whether the systems have the desired effect is part of future work.

4 Discussion

4.1 Comparative Overview

Pepper was released in 2014, which could be a reason for some of the negative key points such as its outdated language model. Talking with the robot about personal problems was also discussed in other studies [23]. Due to the continuous progress in speech recognition, new applications have been developed for Pepper, presenting an interface for using ChatGPT [4]. In the work of Sheridan [19] and Papadopoulos et al. [17], it was found that robots gave ore unsatisfactory answers when asked simple questions. For more complex questions, the robot's performance was similar to that of a human. As noted in [19], collision avoidance receives special attention. This requires not only localization, but also that the robot is aware of its environment. As seen with Pepper, collision avoidance worked well, but localization of the robot in the building had shortcomings. In recent years, Simultaneous Localization and Mapping (SLAM) algorithms have greatly improved through the use of more accurate sensors and more robust algorithms, providing suitable solutions to this problem.

Interaction is affected by further parameters. For example, gestures and reactions are crucial for human-robot interaction [19]. Improving the touch sensors in the hands could not only improve interaction scenarios like shaking hands, but also provide a basis for giving the robot information needed for grasping tasks. Using Pepper as an example, the grasping strength must be enhanced in order to grasp objects. The necessary trajectory planning along with velocity profiles, orientation, and prediction of the target location influence user experience and acceptance of the system [19]. In addition, it has been shown that physical interaction with a soft and warm robot is preferred [19]. In visual design, perception depends on the individual [17]. Insights gained by studying the "uncanny valley" effect showed that a mixture of abstract and human-like appearance can help. As an enabler for social humanoid robots, personalization by adapting to the

Table 2. Functional and non-functional requirements for robot application in assisted work.

No.	Requirement	Description	Funct.	Non-Funct.
1	Collaborative	The robot must stop in the event of unintentional contact with the user [7]	x	
2	Unsupervised use	The robot must perform automated tasks	x	
3	Graphical or manual teach mode	Non-expert users must be able to reprogram the robot for new tasks without extensive training	x	
4	Implementation of Quality control	Cameras and sensor systems must be deployable	x	
5	Mobile setup	The robot must be movable to be used at different work stations	x	
6	Automatic calibration	When the robot is moved to a new task/position, it should be able to recalibrate its base and teach the new position by itself	x	
7	Adaptability in speed	The robot must be adaptable to the speed of the user. The robot should wait in certain positions for the user to take the parts	x	
8	Non-intimidating appearance	Users should not feel threatened by the robot		x
9	Robotic Arm	Unlike other types of robots, the robotic arm provides the necessary flexibility to be used in a variety of environments	x	
10	Robust against high external forces	Rough contact with users cannot be excluded. Therefore, higher external forces must be expected	x	
11	Robust against lighting changes	Quality control cameras need to perform equally well in a variety of lighting conditions due to the mobile nature of the setup	x	

user's tastes and preferences has been mentioned as it is available for Pepper [17].

Additional features such as walking and supervisory tasks were suggested by the caregivers and residents. As the question arose whether a robot would be a suitable walking aid, Karunarathne et al. showed in a study that elderly people would prefer a humanoid robot over walking alone [9]. However, the humanoid robot used lacks a suitable design as a physical aid for people with motor impairments and its inability to move on roads or even on gravel. The question is whether the user would accept the robot over a human companion, and whether cognitively impaired people would react in the same way as the participants in this study. In general, a robot used as a walking aid should be small, equipped with handles, and preferably have a seat and wheels capable of traversing different floor surfaces.

In the workshop environment, the requirements analysis showed differences from general industrial production, which makes the ubiquitous use of robots more important. The focus is on maintaining manual jobs to ensure prospects for people with disabilities. In particular, the robot must be used at different

workstations, assisting people without fixed cycle times, but also adapting to the abilities of each user, resulting in a high level of social behavior. Since the people living in the community housing also work in the company, the crossover effects between social and collaborative robots in the user experience are of interest. No relevant literature was found regarding this topic. It is assumed that there should be no significant drawbacks when using both systems in one facility, since the robotic arm used in the assembly tasks is very different from humanoid robots such as Pepper used in this study (Fig. 3). A challenge in using both robots could be that Pepper actively avoids unintentional contact with the user and its grasp strength is low. On the other hand, industrial and collaborative robots can grasp more strongly and move faster. The user may have difficulty estimating the robot's strength. This could result in the person refusing to cooperate with the robot.

4.2 Ethical Questions

Working with people with cognitive impairments, the institution mentioned that using the robot as a walking aid would improve the quality of life for some residents. In this case, the robot should respect the human's freedom of choice, but also ensure that the environment is safe and that the human does not get into risky situations. This raises the question of whether a robot can take over this care task from a human, and what safety functions need to be integrated to meet these requirements.

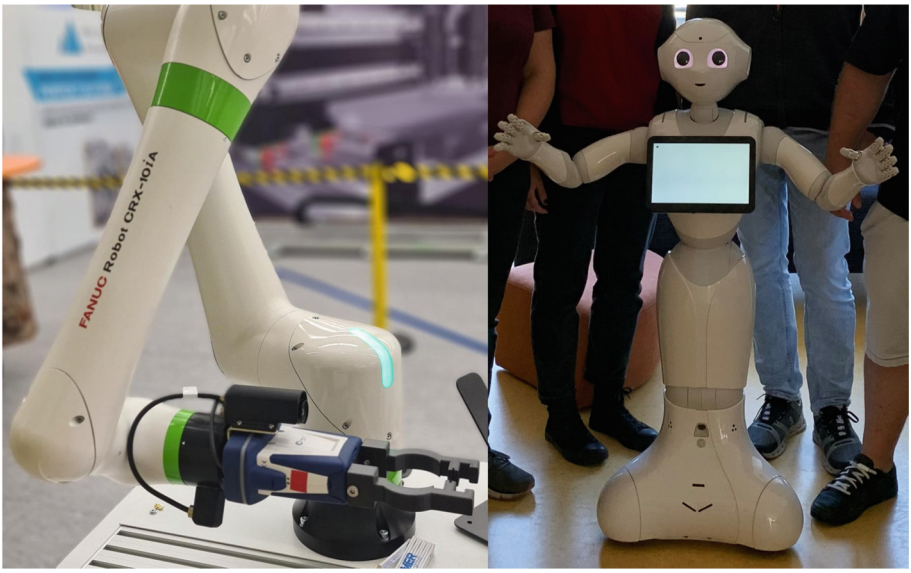

Fig. 3. Use of robots in different environments. Left: Example of a collaborative robot with a camera for assistive work, Right: Social robot Pepper used in assisted living.

In the context of privacy, this work revealed a requirement that contrasts with ethical choices. Including applications that allow the robot to be called by a bed bound person would improve the interaction possibilities between the user and the robot. However, one requirement was that the robot should not enter private rooms. Exceptions need to be defined. In addition, data security is an ongoing issue in robotics due to the use of cameras or the recording and processing of human voice in private environments, especially regarding cloud-based solutions as mentioned above. Camera imagery and LiDAR data have a huge impact due to their detailed imaging of people and environments. This raises concerns in combination with recent legal uncertainties, the increasing autonomy of robots and the replacement of human interaction [3]. With more sophisticated and faster ways to process information, the definition of human-centered data changes. For exoskeletons or adaptive robots, muscle activity detected by the robot or user behavior can be included as part of the human-centered data [3]. To find the right balance between not providing needed functionality due to a potential lack of data and creating a risk to personal security, Fosch-Villaronga et al. suggest involving real users in the process and being transparent about the data collection process so that people can decide for themselves whether they want to share the data [3]. Not only the opinion of the disabled person and caregivers should be sought, but also that of relatives, as in some cases they have a care directive. Including such a number of stakeholders the use of private information should be as clear as possible [8]. The experience from this work shows that each participant (manager, caregiver, and resident) was happy to have this opportunity to help. Nevertheless, it has to be clear that there is a huge responsibility when working with e.g. dementia patients or participants with disabilities [17].

5 Conclusion

This paper presents a case study for social robots in assisted living using the example of Pepper, which was tested for eleven days by residents and caregivers in a community housing complex for assisted living, and for general robotic application and automation in assisted work. For both cases, design and usability requirements were extracted based on user experience. The results were analyzed in comparison with existing literature. The comparison showed similar results. Ethical issues were raised by the organization and the company, which were not found in the literature.

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